



Smart Greenhouse: Automating Agriculture through Closed-Loop Feedback

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ABSTRACT

The purpose of this study is to address Egypt's Grand Challenges of increasing the industrial and agricultural base, pollution reduction, dealing with population growth, and waste recycling by developing a function. Our solution is a smart farming that aims to automate Agriculture through a Closed-Loop Feedback that makes use of sensors and actuators that respond to deviations.

To achieve this, we designed, constructed, and tested a greenhouse that can run uninterrupted by human intervention while maintaining optimal growth conditions for the selected plant species, mushrooms, for more than 6 hours.

The prototype was engineered to meet design requirements, including a quick actuator response, accurate sensor readings, portability, low cost, and the ability to display and visualize data while providing real-time monitoring. Testing included running the prototype for prolonged periods, measuring the area of the growing area, obtaining accurate sensor readings, while maintaining low power consumption.

Major findings through our analysis showed that the prototype can run without human intervention for 6 hours straight, allowing the mushroom to fully sprout. The sensors showed accurate readings over the course of 5 days, while the actuators showed a quick response to deviations. An HTML website is used to display that information on any device.

In conclusion, this project shows how smart greenhouses can increase the industrial and agricultural base of countries, allowing for better environmental and economic conditions, while providing higher crop yields and reducing waste by investing recycled materials into building the greenhouses.

INTRODUCTION

Egypt faces increasing pressure on its agricultural sector due to population growth, environmental degradation, and limited natural resources. These issues are directly linked to Egypt's Grand Challenges, particularly the need to increase the industrial and agricultural base, recycle waste for economic benefit, manage population growth, and reduce pollution in the air, water, and soil. Addressing these challenges requires innovative agricultural systems that are efficient, sustainable, and technologically advanced.

One of the most critical issues threatening agriculture in Egypt is climate change, particularly rising temperatures and their direct impact on crop productivity. As shown in **Figure 1**, rising temperatures are expected to significantly reduce yields of major crops grown in Egypt, highlighting the vulnerability of traditional farming systems to changing environmental conditions. This highlights a clear problem: current agricultural methods are not sufficiently controlled or adaptive to maintain stable and efficient crop production.

Several large-scale solutions have been developed to address these issues, each with notable strengths and limitations.

The WAFRA aquaponic system, developed through international collaboration, demonstrates strong potential through its solar-powered operation, continuous monitoring, and closed-loop design that efficiently uses water and nutrients. However, it requires advanced technical knowledge, has high initial costs, and remains largely in the research phase. (Ain Shams University, n.d.; Kooperation International, n.d.).

Similarly, the NIOF Smart IMTA-Aquaponics system achieves high efficiency in nutrient recycling and integrates multiple species with advanced IoT monitoring, significantly improving productivity and sustainability. Despite this, it is highly complex, expensive to implement, and difficult for small-scale farmers to manage. (VisualStudio 2022, 2024). These examples show that while advanced systems can be highly effective, they often lack accessibility, simplicity, and scalability.

To overcome these limitations, our team established key design and solution requirements. The system must be low-cost and accessible, ensuring it can be widely implemented; simple and replicable, allowing easy construction and modification; and user-friendly, enabling operation without specialized expertise. Additionally, it must support multi-parameter environmental monitoring, measuring factors such as temperature, humidity, and soil moisture, while maintaining automation for extended operation (6 continuous hours) without human intervention. The design must also promote sustainability through the use of recycled materials and avoid generating secondary environmental issues.

Based on these requirements, our team selected a smart greenhouse system utilizing a closed-loop feedback control mechanism. The system continuously monitors environmental conditions through sensors and automatically activates actuators such as irrigation systems, fans, and lighting to maintain optimal plant growth conditions. This ensures consistent regulation of the environment, improving productivity while reducing resource waste.

To implement this solution, appropriate materials and components were selected to ensure accurate sensing, reliable control, and efficient operation. The following section outlines the materials and methods used in constructing the prototype.

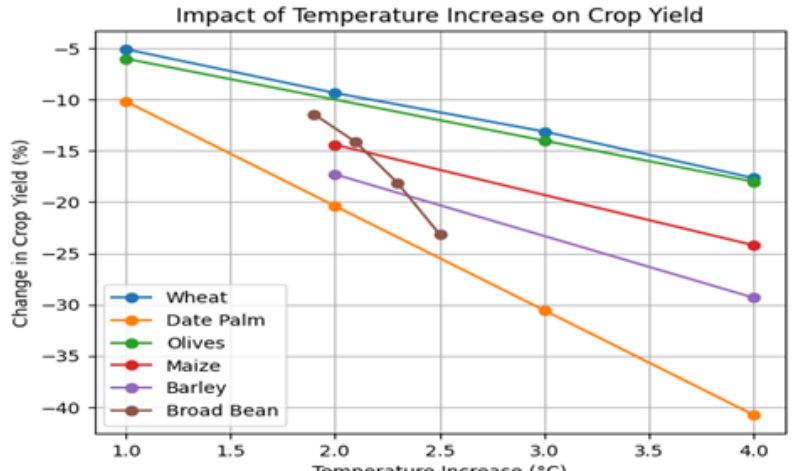


Figure 1: Estimated Reductions in Yield for Specific Crops in Egypt Under Rising Temperature Conditions

METHODS

Phase 1- Building the Greenhouse Structure:

A compact greenhouse frame ($\leq 0.25 \text{ m}^2$) was constructed using lightweight, low-cost materials to ensure portability and ease of replication. Transparent plastic covering was used to allow maximum light penetration while maintaining internal environmental conditions. The full assembled prototype is shown in **Figure 2**.

Phase 2 - Installing Water, Air, and Plant Support Systems:

A simple irrigation system was integrated using tubing and a small water pump to regulate soil moisture. Ventilation openings and airflow pathways were added to control humidity and temperature. Plant holders were positioned to ensure uniform exposure to light and water distribution.

Phase 3 - Electronics and Control System:

The system was built around a microcontroller to process sensor data and control actuators. Sensors for temperature, humidity, soil moisture, and light intensity were connected to continuously monitor environmental conditions. Based on sensor readings, the controller activates actuators such as fans, water pumps, and LED lighting, forming a closed-loop feedback system that maintains optimal conditions automatically. The entire circuit is shown in **Figure 2**.

Phase 4 - Calibration of Sensors:

Each sensor was calibrated to ensure accurate readings. Known reference conditions were used to compare expected and measured values, and correction factors were applied where necessary. The calibration results and relationships between sensor output and real values are presented in **Figure 3**.

Phase 5 - Prepare the Actuators:

Actuators (fan, pump, and LED) were tested individually and then integrated into the system. Trigger thresholds were programmed so that each actuator responds appropriately to environmental changes, ensuring stable and autonomous operation for extended periods.

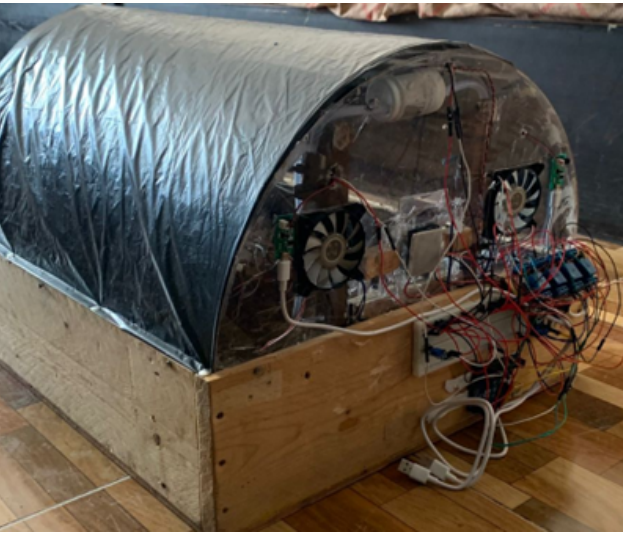


Figure 2: Fully assembled prototype, and circuit.

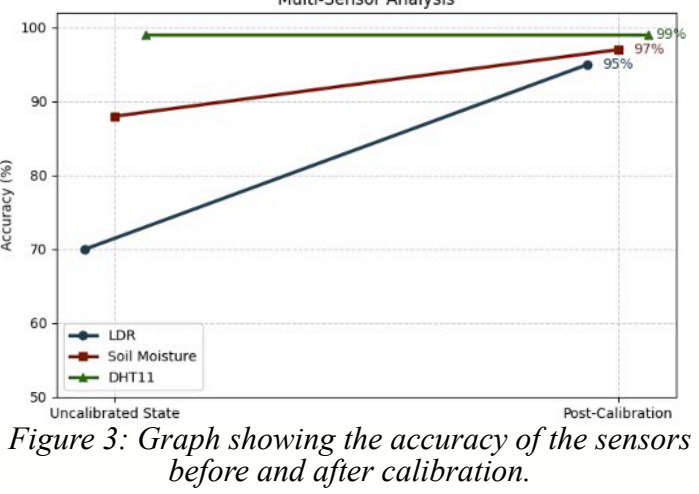


Figure 3: Graph showing the accuracy of the sensors before and after calibration.

TEST PLAN

- 1- Real-time monitoring, data visualization, and functionality:** The prototype must run over six hours, outputting parameter readings and displaying them as graphs. It passes if it maintains operation and correctly shows the data over time.
- 2- Growing Area measurement:** The prototype's growing area must not exceed 0.25 m^2 . It is measured using a measuring tape, and it passes if the calculated size stays within this limit.
- 3- Actuator Implementation and variation from normal conditions:** The system must include actuators that regulate parameters and handle deviations by detecting, alerting, and correcting them. It is tested by introducing disturbances and measuring how quickly the system restores normal conditions.

RESULTS

Negative Results:

Initial testing of the prototype revealed several limitations in the electrical and control systems. The first issue appeared in the lighting system, where a LED strip was initially used. This strip required AC power, while the rest of the system operated on DC, creating incompatibility within the circuit. In addition, the LED strip did not allow proper control over light intensity, reducing the system's ability to regulate light conditions accurately.

A second major issue was observed in the temperature control system. The original H-bridge module used to control the Peltier device did not include a heat sink. Since the Peltier operates by reversing current flow for heating and cooling, the H-bridge experienced rapid overheating during operation, eventually leading to component failure. These issues highlighted the need for better component selection and thermal management in the system design.

Positive Results:

After implementing fixes, the system was retested. The LED strip was replaced with controllable DC LEDs, and the H-bridge was upgraded with a heat sink to prevent overheating. These improvements resolved the initial issues and led to the enhanced performance shown in the positive results. After sensor calibration, the sensors demonstrated consistent and reliable readings over a period of five days under similar environmental conditions, confirming their accuracy and stability, as shown in **Table 2**. All actuators functioned effectively and responded quickly to environmental changes. The cooling system (Peltier module with three fans) successfully reduced the temperature from 26°C to 24°C within 257 seconds. The irrigation system increased soil moisture from 58% to 81% in 47 seconds. The humidifier operated as expected to maintain humidity levels, while the LED lighting system activated instantly when light intensity dropped below 250 lux.

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The prototype successfully operated for 6 continuous hours without human intervention. During this period, sensors continuously monitored environmental conditions, and actuators responded automatically to any detected deviations, confirming the effectiveness of the closed-loop feedback system. The system demonstrated efficient energy usage. Under normal conditions (no actuator activation), the power consumption was approximately 11.6 W, corresponding to a total energy consumption of 250,560 J. This variation is illustrated in **Figure 4**, which shows the relationship between actuator activity and total energy usage.

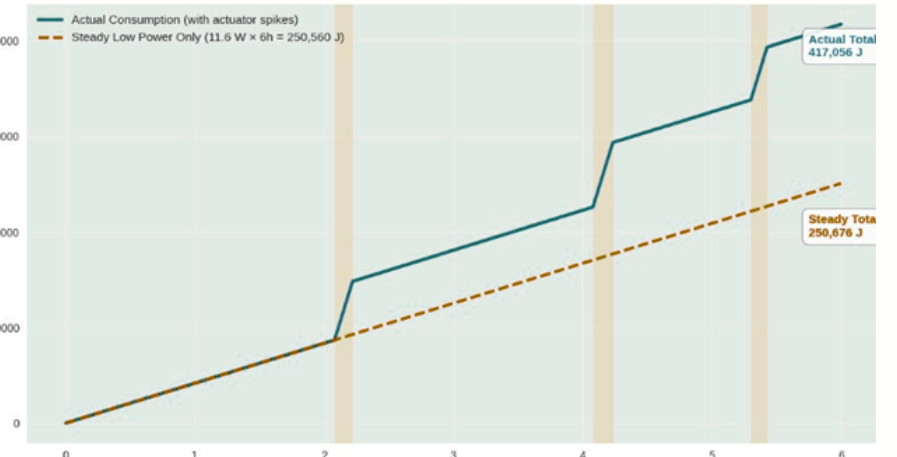


Figure 4: Graph showing the power consumption of the prototype over 6 hours

ANALYSIS

This prototype addresses key challenges in Egypt's Grand Challenges. By tackling two main aspects of the issue of increasing pressure on Egypt's agricultural sector due to population growth and limited resources. The lack of automation agriculture system produces structural inefficiencies across productivity, cost, labor, water, and food security. The prototype offers a sustainable solution that enhances agricultural capacity.

Considering the test results confirm that the prototype meets the design requirements, being fully automated, using recycled material, and multi-parameter monitoring.

The plant that is going to grow in the greenhouse is a mushroom, and it mainly depends on four parameters, which are temperature, humidity, soil moisture, and light intensity.

The first parameter is temperature, which is a fundamental environmental parameter that directly influences biological and chemical processes. Temperature control is critical to maintain stable conditions. The system was designed to maintain a temperature between 18°C and 24°C using a Peltier module with fans. During the test, realistic deviations were introduced by applying a heater as it rose to 26°C , the system detected the variation in 2 seconds and restored it to 24°C in 257 seconds, as shown in **Figure 5**. The thermal resistance of the cooling system is determined by this law

$$R_{th} = \frac{\Delta T}{Q}$$

where ΔT represents the temperature difference, which is 2°C , and the Q represents the power of the Peltier multiplied by the coefficient of performance, which equals $0.6 = 50\text{W}$. The thermal resistance equals $0.04 = \frac{2}{84 \times 0.6}$, which is efficient as it represents how many degrees change for every 1 Watt. This precise control is essential for mushroom cultivation, ensuring protection from rotting and supporting high-quality growth.

The second parameter is Humidity, which is the concentration of water vapor in the air. As it directly affects transpiration, water uptake in the plants. From the results, the system maintained the ideal range for the humidity, which is between 60% to 80%. Regulating humidity is important to protect the mushroom from fluctuations, as the system uses two active humidifiers if the humidity goes out of the ideal range.

The third parameter, which is Soil moisture, is the amount of water present in the soil expressed as a percentage, measured by the YL-69 sensor that uses Ohm's law.

(Ohm's Law - University Physics Volume 2 - OpenStax, 2016b) As the sensor applies a fixed voltage, it measures the resulting current and measures resistance through Ohm's law to determine if the soil is wet or dry. The system targets a range of 60% to 80% to maintain a healthy life for the plant. For testing the soil moisture, dry soil was used; the system detected the deviation in 2 seconds, and the system restored the deviation by applying the pump and restored it to 80% in roughly 47 seconds, as shown in **Figure 6**.

The fourth parameter, which is Light intensity, is the amount of light energy falling on a surface, as it is essential in agriculture because it controls photosynthesis and plant development. Different plants need different wavelengths for their development, as different wavelengths affect different biological systems differently. The system uses a Light-Dependent Resistor (LDR) to measure light and switches on/off the lamps immediately, as the sensor is based on photoconductivity.

From obtaining the tests, these Parameters remained in their setpoints most of the time in the six-hour test without human intervention, maintaining a healthy environment for germination of mushrooms.

Sensors have variability in readings caused by inconsistencies between the sensors, which increased the error margin by more than 20%. This issue was resolved by implementing **(MA.2.01)**. This involved establishing a linear function for prototype calibration, utilizing the standard linear function for this purpose: $Y = mx + c$. Solving the constants until the prototype is calibrated, taking reference points from an example hygrometer in measuring temperature and humidity as follows.

$R_{cal} = \text{Calibration Scale} \cdot R_{raw} + \text{Calibration Offset}$

This mathematical correction reduced the error value of the LDR as shown in **Figure 4** by nearly 83%, as shown in the following equation by applying **(PH.1.01)**.

$$\text{Error Reduction} = \frac{(\text{Uncalibrated Error} - \text{Calibrated Error})}{\text{Uncalibrated Error}} = \frac{83.33\% = \frac{(0.30 - 0.05)}{0.30}}$$

In the initial testing, there were negative results. An H-bridge without a heatsink was used, and because it uses four switches to forward and reverse the Peltier direction, as it is based on the Lorentz force, Back EMF produces heat, so it leads to burning the H-bridge, and it is replaced with one with a heatsink so it can withstand heat.

In the initial testing, the light strip wasn't connected directly in the circuit, as it needed an AC connection, along with consuming lots of power, so it was replaced with a grow light for more efficient lighting.

The actuators are the electronics that are responsible for restoring the deviations happens in the system and the application of DC motors that convert electrical energy into mechanical rotation, which work by electromagnetism based on the Lorentz Force law. When current flows through a coil placed in a magnetic field, the coil experiences a force that rotates it $F \propto I$ **(PH.2.07)**, and a commutator is placed to reverse the current in every half direction to keep the rotation continuous **(PH.2.10)** (Horowitz, P., & Hill, W. (2015).

The Maximum power calculated for the prototype by performing a summation of the peak power requirements of all electric components

$$P_{total} = \sum P_{component}$$

, with all the actuators operating simultaneously. Peltier (84W) + 2 humidifiers (4W) + 3 cooling fans (6.12W) + 2 water pumps (16.8W) + 4 relays (3.7W) + H-bridge(5W) + sensors (0.3W) + Arduino uno (2.5W) all equals 122.52W. It is calculated through the power formula, which is derived from Ohm's laws. $P = V \times I$ **(ME.2.05)**.

The system remains efficient by activating actuators only when needed, reducing overall consumption.

CONCLUSION

The problem addressed in this project was the challenge of improving agricultural productivity in Egypt under changing environmental conditions, while maintaining low cost, sustainability, and accessibility. This directly aligns with Egypt's Grand Challenges, particularly increasing the agricultural base and reducing environmental impact. The aim was to develop a small-scale smart greenhouse capable of automatically regulating key environmental parameters using a closed-loop feedback system.

Through prototype construction and testing, the system successfully monitored temperature, humidity, soil moisture, and light intensity, and responded using actuators without human intervention for 6 continuous hours. The results confirmed that the prototype met its main design requirements: it is low-cost, automated, energy-efficient, and capable of maintaining stable growing conditions. The quick response times of actuators and consistent sensor readings validate the effectiveness of the control system.

Compared to prior solutions such as the WAFRA system and the NIOF Smart IMTA-Aquaponics system, our prototype offers a more accessible and simplified approach. While those systems achieve high efficiency and advanced monitoring, they require significant financial investment, technical expertise, and complex infrastructure. In contrast, our solution focuses on automation, affordability, and ease of replication, making it more suitable for small-scale farmers, while still maintaining the core benefit of automated environmental control.

Therefore, it can be concluded that the proposed smart greenhouse system is a practical and scalable solution that balances efficiency with accessibility. It demonstrates that automated agriculture does not necessarily require complex or expensive systems, and that simpler designs can still provide meaningful contributions toward sustainable agricultural development in Egypt.

RECOMMENDATIONS

After completing the project, the prototype fulfilled all the design requirements, and the project produced effective and great results. However, there are certain suggestions that must be incorporated if the project is carried out in the future to increase its effectiveness and yield better results.

One key improvement is increasing the level of control over the system by adding relays, as it will allow the fans to operate independently rather than as a grouped mechanism, enabling more precise regulation. Expanding the prototype on a large-scale automated agricultural house would improve its practical applicability. Integrating solar panels as the power source for our prototype will improve efficiency.

Further enhancements in terms of learning machines by incorporating a machine learning algorithm trained on data and trials to allow the system to make decisions like humans. Additionally, connecting the Arduino to an API to a database so the system will get continuous updates on the parameters, ensuring the system remains accurate if there is a change in the ideal conditions the plant lives in

To protect the prototype, implementing a fuse would protect the system from overcurrent, in addition to implementing a cooling mechanism for the electronics to prevent them from being burned and damaged. Finally, adding a backup power system, such as batteries, would ensure continuous operation through power outages, increasing reliability.

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MATERIALS

Acrylic & Wood Recycled Main building materials of the prototype, the wood was used for the base and main structure, while the acrylic was used for covering the structure of the prototype.	Arduino-UNO 450 LE A low-cost, energy-efficient microcontroller, designed for IoT and embedded applications.
Breadboard Reused A breadboard is a solderless platform to connect everything to.	DHT-11 50 LE Used to measure temperature and humidity, providing data in a digital format.
Fans & Humidifiers Reused The Fan is used to lower the temperature by replacing hot air with cold air, and distributing droplets of the humidifier, while the humidifier adds moisture to the air to regulate humidity.	LDR-Module 35 LE Is a sensor whose electrical resistance decreases as light intensity increases, allowing it to detect and measure brightness
Peltier Reused A thermoelectric device that transfers heat from one side to the other when electricity flows, enabling cooling or heating without moving parts	Relays 40 LE x3 A pre-built circuit that includes a relay and control components, allowing the microcontroller to safely turn high-power devices on or off
Soil Moisture 40 LE Measures the water content in soil, helping optimize irrigation, plant health, and environmental monitoring	Water Pump & Hose Reused Moves water by converting electrical energy into fluid motion

Table 1: Materials used in the construction of the prototype Total Cost: 695